CNN在不同任务上的应用

Playing Go

Network

Next move
(19 x 19 个
位置)

19 x 19 vector

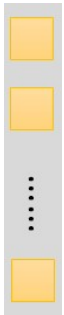
黑子: 1
白子: -1
空: 0

可以用全连接网络

但CNN的表现更好

Playing Go

Training: record of previous plays 黑: 5之五 → 白: 天元 → 黑: 五之5 ...



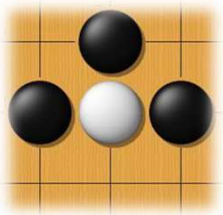
Target:
"天元" = 1
else = 0

Target:
"五之 5" = 1
else = 0

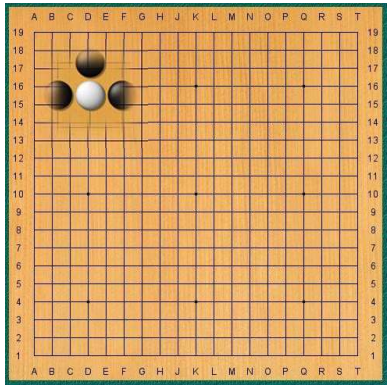
Why CNN for playing Go?

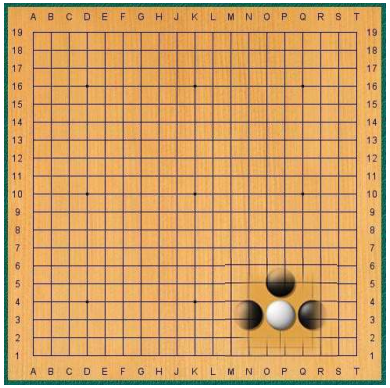
- 有些模式比整个图像小得多

  Alpha Go 的第一层用 5 x 5 的卷积核



- 相同的模式出现在不同的区域中。





Why CNN for playing Go?

- 对图像进行二次采样不会改变图像中的物体

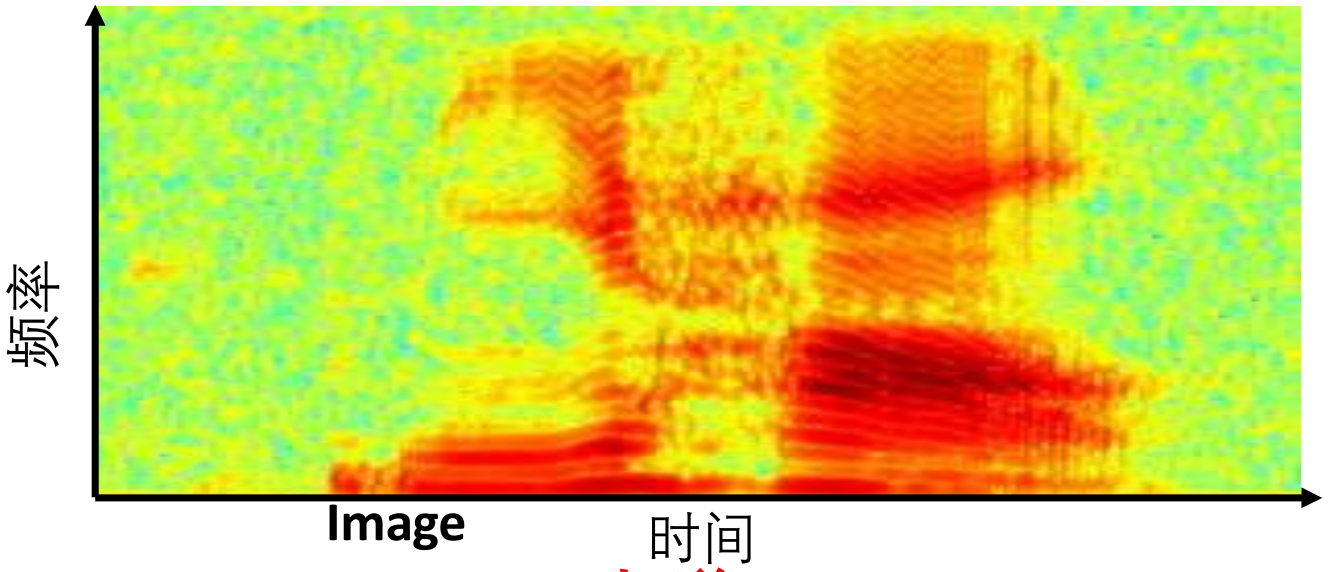
Max Pooling

How to explain this???

Neural network architecture. The input to the policy network is a $19 \times 19 \times 48$ image stack consisting of 48 feature planes. The first hidden layer zero pads the input into a 23×23 image, then convolves k filters of kernel size 5×5 with stride 1 with the input image and applies a rectifier nonlinearity. Each of the subsequent hidden layers 2 to 12 zero pads the respective previous hidden layer into a 21×21 image, then convolves k filters of kernel size 3×3 with stride 1, again followed by a rectifier nonlinearity. The final layer convolves 1 filter of kernel size 1×1 with stride 1, with a different bias for each position, and applies a softmax function. The [illegible] Extended Data Table 3 additionally show the results of training with $k = 128$, 256 and 384 filters.

Alpha Go does not use Max Pooling .....

语音识别



频谱图

语音识别

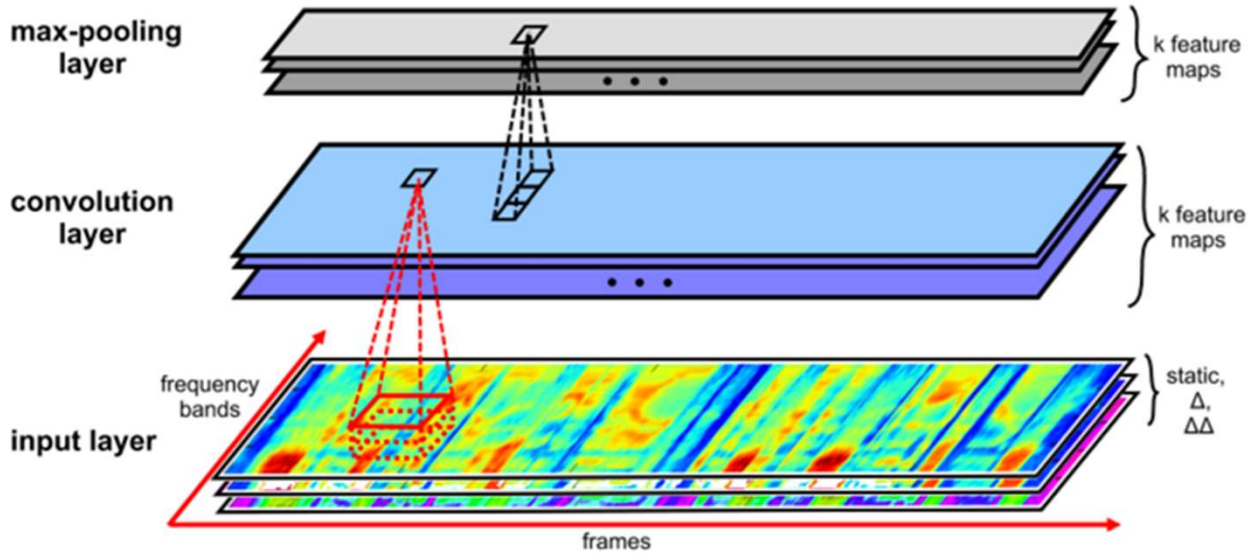


Figure 1: *The convolution layer and max-pooling layer applied upon input features.*

[1701.02720] Towards End-to-End Speech Recognition with Deep Convolutional Neural Networks (arxiv.org)

语音识别

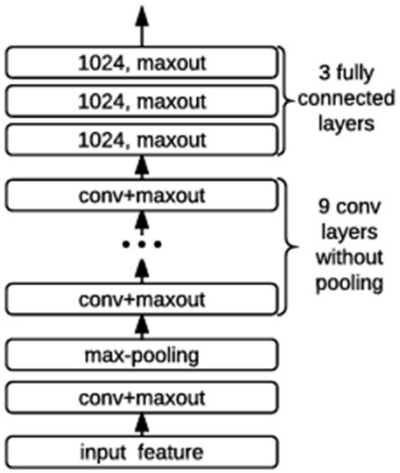
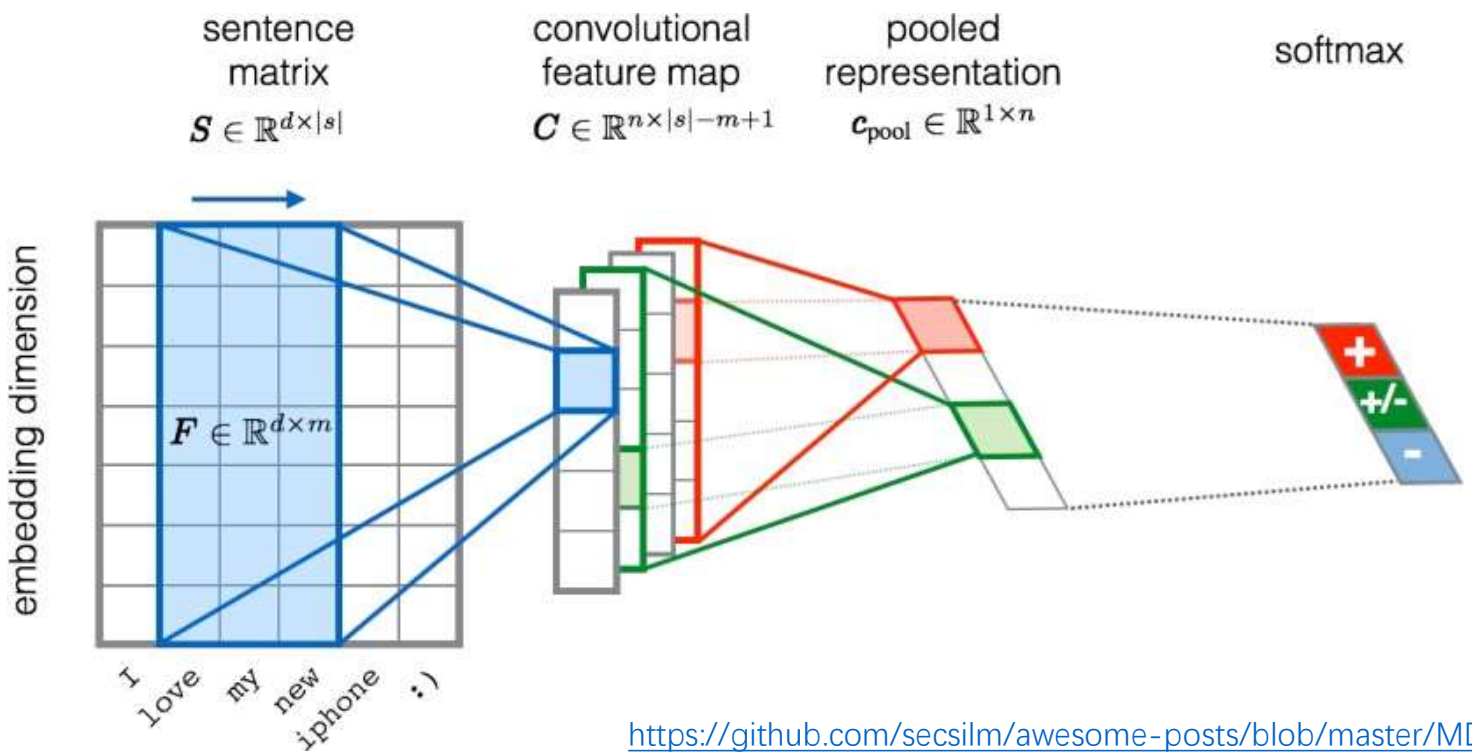


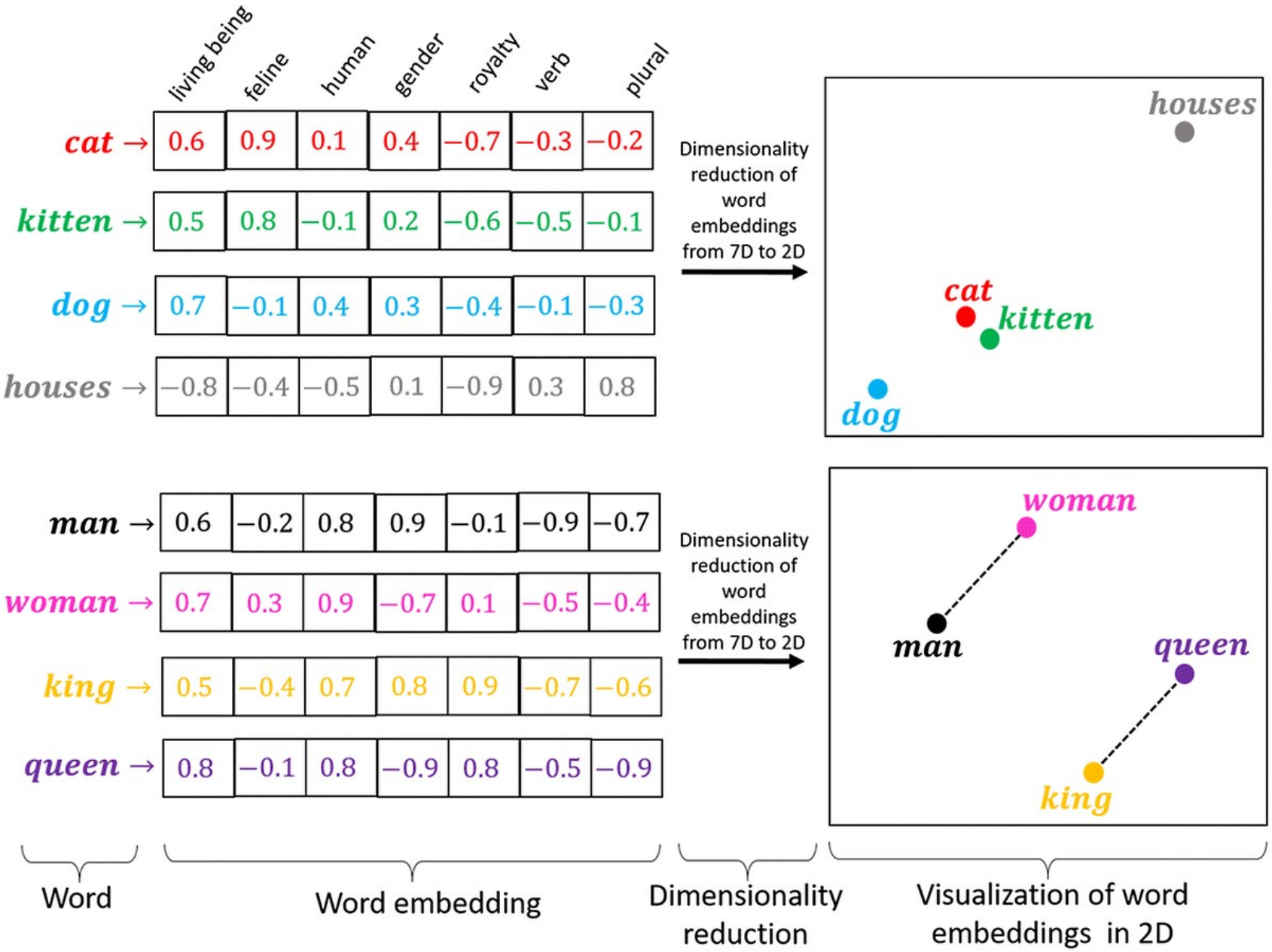
Figure 3: *Network structure for phoneme recognition on the TIMIT dataset. The model consists of 10 convolutional layers followed by 3 fully-connected layers on the top. All convolutional layers have the filter size of* 3×5 *and we use max-pooling with size of* 3×1 *only after the first convolutional layer. First and second numbers correspond to frequency and time axes respectively.*

[1701.02720] Towards End-to-End Speech Recognition with Deep Convolutional Neural Networks (arxiv.org)

文本分类



https://github.com/secsilm/awesome-posts/blob/master/MDs/text-classification-with-tensorflow-estimator.md


living being
feline
human
gender
royalty
verb
plural
cat → 0.6 0.9 0.1 0.4 −0.7 −0.3 −0.2
kitten → 0.5 0.8 −0.1 0.2 −0.6 −0.5 −0.1
dog → 0.7 −0.1 0.4 0.3 −0.4 −0.1 −0.3
houses → −0.8 −0.4 −0.5 0.1 −0.9 0.3 0.8
Dimensionality reduction of word embeddings from 7D to 2D
houses
cat
kitten
dog
man → 0.6 −0.2 0.8 0.9 −0.1 −0.9 −0.7
woman → 0.7 0.3 0.9 −0.7 0.1 −0.5 −0.4
king → 0.5 −0.4 0.7 0.8 0.9 −0.7 −0.6
queen → 0.8 −0.1 0.8 −0.9 0.8 −0.5 −0.9
Dimensionality reduction of word embeddings from 7D to 2D
woman
man
queen
king
Word
Word embedding
Dimensionality reduction
Visualization of word embeddings in 2D